



BEST PRACTICE SERIES

8 Step Preventive Maintenance Process

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

8 Step Preventive Maintenance Process 0

1.0 Introduction 1

2.0 Best Practice Description..... 1

3.0 Implementation Steps 3

4.0 Benefits..... 5

5.0 Resources Required 6

6.0 Supporting Attachments / References 6

7.0 Related Best Practices 6

8.0 Acknowledgements..... 6

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.



1.0 Introduction

Preventive Maintenance (PM) is critical to the success of any maintenance and repair strategy. It is one of the foundation elements and should be considered one of the focal points in the overall equipment maintenance management strategy. The goals of PM are to maintain equipment in a productive and safe operating condition, to reduce the risk of failure and subsequent unscheduled equipment repairs, and to optimize equipment owning and operating costs.

This Best Practice describes the 8 Step Preventive Maintenance Methodology. It supports the Preventive Maintenance Best Practice (1007-2.0-1101) and is an alternative to what is known as the 4 Step or Traditional PM Methodology.

Inside of Global Mining's "maintenance and repair model", the direction is for a maintenance model of "condition based maintenance". In other words Caterpillar would like to see a "repair before failure" methodology. To support this, an overall maintenance and repair strategy should include the processes of condition monitoring, backlog management, and planning and scheduling.

The 8 Step PM Methodology enables opportunity for the execution of planned and scheduled backlogs at every PM service. From experience, the authors have seen that the smallest and longest duration PM services do not always have backlogs scheduled at those times due to a desire to keep an acceptable level of availability. What has been found in a "repair before failure" model is by ensuring an opportunity to schedule and complete backlogs at every PM service provides the best opportunity to maximize availability and reliability while minimizing costs.

2.0 Best Practice Description

A Traditional PM service schedule (PM 1, 2, 3 & 4 methodology – 250, 500, 1000 and 2000 hours) results in service durations that vary based on the number of tasks that are required to be undertaken. As an example, a 250 hour PM with a small amount of tasks can have a downtime duration of 4 hours. Conversely, a 2000 hour PM where every maintenance task, oil and filters change is carried out, the downtime can be as much as 24 hours (or more) depending on the machine type, labor, and resources available.

An 8 Step PM Methodology distributes all of the Preventive Maintenance tasks across the service intervals in a 2000 hour period in order to achieve PM services of equal downtime duration.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

8 Step Preventive Maintenance Process	DATE 6/17/2011	CHG NO 01	NUMBER 1011-1.01-1219
---------------------------------------	-------------------	-----------------	--------------------------

The 8 Step vs. 4 Step Methodology is summarized in Figure 2.1 below:

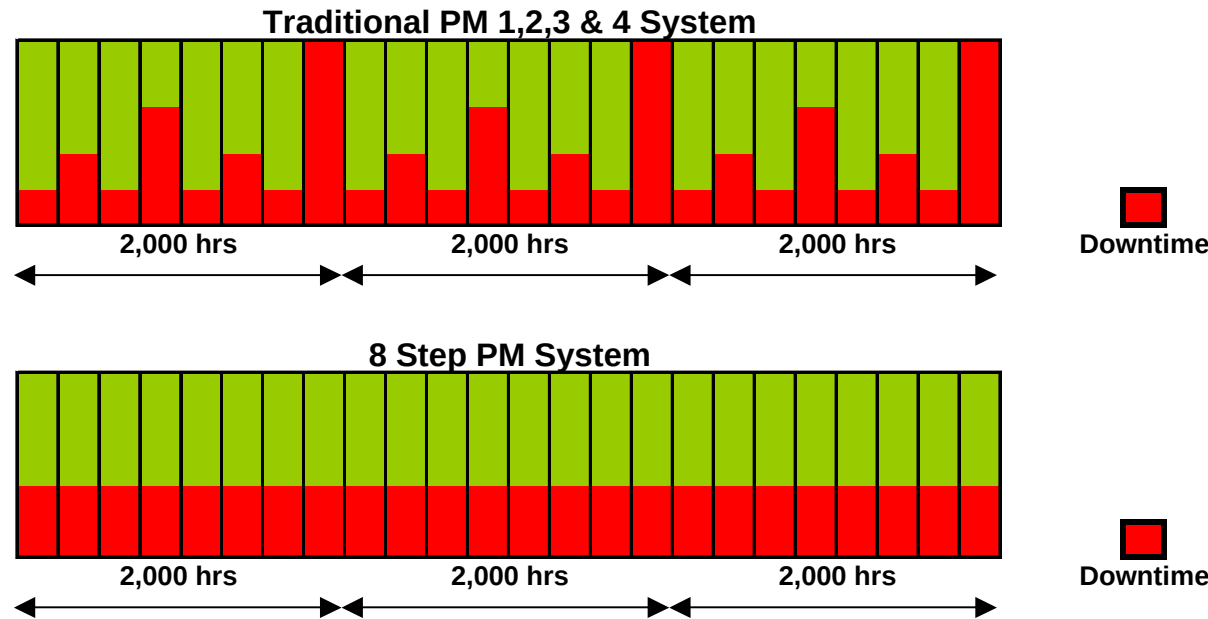


Figure 2.1: Traditional and 8 Step PM Process

No tasks or their intervals differ in the 8 Step Process from those of the traditional (4 Step) services. An explanation of this will be shown in the implementation steps below.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
8 Step Preventive Maintenance Process	DATE 6/17/2011	CHG NO 01	NUMBER 1011-1.01-1219

3.0 Implementation Steps

There may be various methods of implementing the 8 Step PM Process. Execution of the process may best be carried out from new or in conjunction with the next 2,000 hour PM.

3.1 Time and Task Redistribution (Example)

		Traditional				8 Step Blend							
		250	500	1000	2000	Fixed 250 hr interval - Wash on every service Blend 250 / 500 / 1000 / 2000 hr service items.							
Interval		PM 1	PM 2	PM 3	PM 4	PM 1	PM 2	PM 3	PM 4	PM 5	PM 6	PM 7	PM 8
Oils													
Engine Oil - Change (ORS?)	500		X	X	X		X		X		X		X
Front Wheel Oil - Change	500		X	X	X	X		X		X		X	
Torque Converter and Trans Oil - Change	1000			X	X			X					X
Engine Coupling Oil - Change	2000				X	X							
Hydraulic Tank Oil - Change	2000				X			X					
Steering System Oil - Change	2000				X				X				
Differential and Final Drive Oil - Change	2000				X						X		
Filters													
Brake Actuation Oil Filter - Replace	500		X	X	X	X		X		X		X	
Brake Cooling Oil Filter - Replace	500		X	X	X	X		X		X		X	
Differential and Final Drive Oil Filter - Replace	500		X	X	X	X		X		X		X	
Engine Coupling Oil Filter - Replace	500		X	X	X		X		X		X		X
Engine Oil Filter - Change	500		X	X	X		X		X		X		X
Fuel System Priming Filter - Replace	500		X	X	X		X		X		X		X
Fuel System Secondary Filter - Replace	500		X	X	X		X		X		X		X
Pump and Motor Case Drain Oil Filter - Replace	500		X	X	X		X		X		X		X
Steering System Oil Filter - Replace	500		X	X	X		X		X		X		X
Torque Converter Oil Filter - Replace	500		X	X	X		X		X		X		X
Transmission Oil Filter - Replace	500		X	X	X		X		X		X		X
Engine Air Filter Primary Element - Clean/Replace	2000												
Based on Condition													

Figure 3.1: This is a demonstration of a set of scheduled oil and filter change tasks. It consists of both the traditional system and the 8 Step Process from which the scheduled oil tasks will be transferred to.

Figure 3.1 demonstrates task redistribution from the Traditional 4 Step System to the 8 Step Methodology for oil and filter changes. The service regime in this example is based on a 250-hour interval over 2000 hours for a fleet of 797B trucks.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

8 Step Preventive Maintenance Process

DATE
6/17/2011

CHG
NO
01

NUMBER
1011-1.01-1219

		Traditional				8 Step Blend							
		250	500	1000	2000	250	500	750	1000	1250	1500	1750	2000
Interval		PM 1	PM 2	PM 3	PM 4	PM 1	PM 2	PM 3	PM 4	PM 5	PM 6	PM 7	PM 8
Oils													
Engine Oil - Change (ORS?)	500		x	x	x		20		20		20		20
Front Wheel Oil - Change	500		x	x	x	15		15		15		15	
Torque Converter and Trans Oil - Change	1000			x	x				20				20
Engine Coupling Oil - Change	2000				x		15						
Hydraulic Tank Oil - Change	2000				x			35					
Steering System Oil - Change	2000				x				10				
Differential and Final Drive Oil - Change	2000				x							40	
Filters													
Brake Actuation Oil Filter - Replace	500		x	x	x	10		10		10		10	
Brake Cooling Oil Filter - Replace	500		x	x	x	15		15		15		15	
Differential and Final Drive Oil Filter - Replace	500		x	x	x	10		10		10		10	
Engine Coupling Oil Filter - Replace	500		x	x	x		10		10		10		10
Engine Oil Filter - Change	500		x	x	x		10		10		10		10
Fuel System Priming Filter - Replace	500		x	x	x		5		5		5		5
Fuel System Secondary Filter - Replace	500		x	x	x		5		5		5		5
Pump and Motor Case Drain Oil Filter - Replace	500		x	x	x	5		5		5		5	
Steering System Oil Filter - Replace	500		x	x	x	5		5		5		5	
Torque Converter Oil Filter - Replace	500		x	x	x		10		10		10		10
Transmission Oil Filter - Replace	500		x	x	x		15		15		15		15
Engine Air Filter Primary Element - Clean/Replace	2000					Based on Condition							
Totals Average						60	90	95	95	70	75	100	95
						85							

Figure 3.2: This is the same example as Figure 3.1 but with the distribution of tasks being allocated with their approximate amount of minutes for completion.

The crosses (x) in the Traditional System columns represent the tasks to be completed for each service. The numbers in the 8 Step Process columns represent the approximate time (minutes) taken to complete the task.

The key is to redistribute the tasks without affecting the task interval and equalizing the overall time taken for each service in 8 Step Process. The exercise should be repeated for all lube, maintenance and mechanical inspection (preventive maintenance) tasks. Once all tasks have been redistributed, sum up the total time taken and ensure the duration for each Step of service is roughly the same. An example is shown Figure 3.2 above.

The 8 Step Process can also be applied to 500-hour interval over 4000 hours for any fleet of machines. However, the maintenance organization should consider the following factors when determining service interval:

- OEM maintenance recommendations
- Overall maintenance strategy
- The size and number of machines in the fleet
- Labor requirements
- On-site tooling and infrastructure
- Site conditions

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
8 Step Preventive Maintenance Process	DATE 6/17/2011	CHG NO 01	NUMBER 1011-1.01-1219

3.2 Change Management

The role of a process owner is critical to a successful transition. He/she should be assigned with the authority and budget to facilitate the implementation of the program. Their responsibilities should include the education and communication to the relevant stakeholders to ensure their support in the 8 Step Process implementation. Stakeholders include positions such as; Maintenance Superintendent, Planner, Scheduler, Maintenance Supervisor, PM Supervisor, Fleet Analyst, Technicians, Technical Communicator and the Production Department.

What has been found from previous experience is to engage key stakeholders early in the process so that everyone within the organization understands the reasons and benefits. To ensure buy-in and ownership from all affected parties it is best to utilize team members from all levels of the organization when defining the process and building the checklists.

4.0 Benefits

The purpose and benefits of the 8 Step Preventive Maintenance (PM) Process are as follows:

- Improve PM efficiency.
- Opportunity to plan and schedule efficient and effective backlog completion at every PM.
- Reduce machine down time through improved machine and fleet reliability.
- Improved planning and scheduling of the PM plan.
- Improved production planning for scheduled PM downtime.
- One crew in one shift can complete the PM service, improving accountability and loss of time over shift changes etc.
- Dedicated and consistent number of people required for each service.
- Ability to schedule consistent number of PM's per shift
 - o i.e. Two PM's per day, one each shift

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

8 Step Preventive Maintenance Process	DATE 6/17/2011	CHG NO 01	NUMBER 1011-1.01-1219
---------------------------------------	-------------------	-----------------	--------------------------

5.0 Resources Required

Please refer to Best Practice “1007-2.0-1101 – Preventive Maintenance Strategy” for basic resources requirement

6.0 Supporting Attachments / References

- 8-Step vs 4-Step PM Calculator.xls
- 8-Steps PM Program.ppt
- 785C 8 Step info.xls
- PM Schedule test.xls
- 8 Step Master.xls

7.0 Related Best Practices

Best Practice 1007-2.0-1101 – Preventive Maintenance Strategy

Best Practice 1007-2.0-1100 – Maintenance and Repair Processes

Best Practice 0607-2.9-1081 – Improved PM Operations Using Specialized Bay

8.0 Acknowledgements

This 8 Step PM Process Best Practice was created by:

Rex Cheung
Development Engineer
Caterpillar Global Mining
+613 9953 9962
Cheung_Rex@cat.com

Please Contact the author for further details or information regarding this Best Practice.

Special mention must be made of Caterpillar Global Mining, WesTrac and Hastings Deering for their willingness to share this Best Practice.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

8 Step Preventive Maintenance Process	DATE 6/17/2011	CHG NO 01	NUMBER 1011-1.01-1219
---------------------------------------	-------------------	-----------------	--------------------------